

REVISED PROGRAM OF STUDY



North Eastern Mindanao State University (NEMSU)
Bachelor of Science in Computer Science (BSCS)
Based on CMO No.25, Series of 2015
Effectivity: 1st Semester AY 2026-2027



2/16/26
DATE

Program of Study

FIRST YEAR - First Semester					
Course Code	Descriptive Title	No. of Hours		Credit Units	Pre-requisite(s)
		Lec	Lab		
CS 111	Introduction to Computing	2	3	3	None
CS 112	Fundamentals of Programming	2	3	3	None
GE-US	Understanding the Self	3		3	None
GE-MMW	Mathematics in the Modern World	3		3	None
GE-PC	Purposive Communication	3		3	None
MATH 1	Advance College Algebra with Trigonometry	3		3	None
IT 1	Living in the IT Era	3		3	None
PATH- Fit 1	Movement Competency Training	2		2	None
NSTP 1	National Service Training Program 1	3		3	
TOTAL		24	6	26	

FIRST YEAR - Second Semester					
Course Code	Descriptive Title	No. of Hours		Credit Units	Pre-requisite(s)
		Lec	Lab		
MATH2	Calculus 1 (Differential calculus)	3		3	MATH1
CS 122	Intermediate Programming	2	3	3	CS 112
CS 123	Programming Languages	2	3	3	CS 111, CS 112
CS 124	Data Structures and Algorithms	2	3	3	CS 111, CS 112
GE-STIS	Science Technology and Society	3		3	
GE-E	Ethics	3		3	
GE-CW	The Contemporary World	3		3	
PATH- Fit 2	Exercise -based Fitness Activities	2		2	PATH- Fit 1
NSTP 2	National Service Training Program 2	3		3	NSTP 1
TOTAL		23	9	26	

SECOND YEAR - First Semester					
Course Code	Descriptive Title	No. of Hours		Credit Units	Pre-requisite(s)
		Lec	Lab		
MATH3	Calculus 2 (Integral calculus)	3		3	MATH2
CS 211	Object-Oriented Programming	2	3	3	CS 122
CS 212	Mobile Computing 1	2	3	3	CS 122
CS 213	Computer Architecture & Organization	2	3	3	CS 124
CS 214	Algorithms and Complexity	2	3	3	CS 124
GE-RVA	Reading Visual Arts	3		3	
GE- AA	Art Appreciation	3		3	
PATH- Fit 3	Menu of Dance, Sports, Martial Arts, Group Exercise , Outdoor and Adventure Activities	2		2	PATH- Fit 1&2
TOTAL		19	12	23	

SECOND YEAR - Second Semester					
Course Code	Descriptive Title	No. of Hours		Credit Units	Pre-requisite(s)
		Lec	Lab		
CS 221	Discrete Structures	3		3	MATH1, MATH2
CS 222	Information Management	2	3	3	CS 211
CS 223	Mobile Computing 2	2	3	3	CS 212
CS 224	Computational Science	3		3	CS 211
Ecos 1	People and the Earth's Ecosystem	3		3	
GE- RPH	Readings in the Philippine History	3		3	
PATH- Fit 4	Menu of Dance, Sports, Martial Arts, Group Exercise , Outdoor and Adventure Activities	2		3	PATH- Fit 1,2&3
CS 225	Human Computer Interaction	2	3	3	CS 213
TOTAL		20	9	24	

THIRD YEAR - First Semester					
Course Code	Descriptive Title	No. of Hours		Credit Units	Pre-requisite(s)
		Lec	Lab		
CS 311	Automata Theory & Formal Languages	3		3	CS 221
CS 312	Information Assurance and Security	2	3	3	CS 222
CS 313	CS Elective 1	2	3	3	CS 222
CS 314	Data Mining Concepts and Techniques	2	3	3	CS 221

CS 315	Application Dev't & Emerging Technologies	2	3	3	CS 222
CS 316	Machine Learning	2	3	3	CS 122&221
GE-E	GE Elective 1 - Gender and Society	3		3	
TOTAL		16	15	21	

THIRD YEAR - Second Semester					
Course Code	Descriptive Title	No. of Hours		Credit Units	Pre-requisite(s)
		Lec	Lab		
CS 321	Software Engineering 1	2	3	3	CS 211&222
CS 322	Thesis 1	2	3	3	Regular 3rd Yr.
CS 323	Technopreneurship	2	3	3	CS 315
CS 324	Operating Systems	2	3	3	CS 123
CS 325	CS Elective 2	2	3	3	CS 122
RIZAL	Life and Works of Rizal	3		3	
GE- Stat	Descriptive and Inferential Statistics	3		3	
TOTAL		16	15	21	

MID YEAR					
Course Code	Descriptive Title	No. of Hours		Credit Units	Pre-requisite(s)
		Lec	Lab		
CS 331	Networks and Communication	2	3	3	Regular 3rd Yr.
TOTAL		2	3	3	

FOURTH YEAR - First Semester					
Course Code	Descriptive Title	No. of Hours		Credit Units	Pre-requisite(s)
		Lec	Lab		
CS 411	Software Engineering 2	2	3	3	CS 322
CS 412	CS Elective 3	2	3	3	CS 314, CS 322
CS 413	Social Issues and Professional Practice	2	3	3	CS 312
GE- BC	Business Correspondence	3		3	
CS 414	Thesis 2	2	3	3	CS 323
TOTAL		11	12	15	

FOURTH YEAR - Second Semester					
Course Code	Descriptive Title	No. of Hours		Credit Units	Pre-requisite(s)
		Lec	Lab		
CS 421	OJT (486HRS)		9	9	Completed Academic Requirements
TOTAL			9	9	

II. Summary

Name of Program: Bachelor of Science in Computer Science PSG: CMO No.25, S. 2015

Courses	Number of Units	Courses	Number of Units
Required by PSG		This Revised Curriculum	
General Education Courses	54	General Education Courses	42
Common Courses	18	Common Courses	21
Professional Courses	48	Professional Courses	75
Professional Electives	9	Professional Electives	9
Additional Math Requirement	3	Additional Math Requirement	9
PE	8	PE	8
NSTP	6	NSTP	6
Minimum Total Units	146	Total	170

Evaluated by: _____

Breakdown of Courses:

Courses	Units
General Education Courses	42
Common Courses	21
Professional Courses	75
CS Electives	9
Math Electives	9

CHED CARAGA

BUTUAN CITY

CONTENTS

AMY C. SANCHEZ, MSCA

Education Supervisor II

2/16/24

DATE

PE	8
NSTP	6
TOTAL	170

Reviewed and Noted by:

AMY C. SAAVEDRA, MSCA

CHED Caraga Regional Supervisor for Information Technology Education

Bachelor of Science in Computer Science (BSCS)	
Elective Courses	
Course Code	Course Description
Third Year- 1 st Semester	
CS 313	CS Elective 1 -Intelligent Agents/IoT
Third Year- 2 nd Semester	
CS 325	CS Elective 2 - Computer Graphics
Fourth Year -1 st Semester	
CS 412	CS Elective 3 - Data Science